Recovery from Deadlock

1 Main Idea

Deadlock Recovery

Deadlock recovery is the action taken after a deadlock has already been detected, so that the system can make progress again.

Detection only tells us that a deadlock exists. Recovery decides how to break it. The common methods are:

- recover through preemption,
- recover through rollback,
- recover by killing one or more processes.

Important

Deadlock recovery methods are usually unpleasant. They may lose work, require manual intervention, or terminate processes.

2 Recovery through Preemption

Preemption

Recovery through preemption means temporarily taking a resource away from one process and giving it to another process so the deadlock can be broken.

This works only when the resource can be removed without corrupting the process or the resource state.

Example: Printer

An operator may pause a process using a printer, collect the printed pages, suspend the process, give the printer to another process, and later restart the original process.

Issue	Explanation
Resource-dependent	Some resources can be taken away safely, but many cannot.
May need manual help	Batch systems may require an operator to intervene.
Victim choice matters	Choose a process whose resources can be taken back most easily.
Often difficult	The process should ideally not notice that the resource was removed.

Limitation

Preemption is often impossible for nonpreemptable resources. For example, taking away a device in the middle of an irreversible operation may corrupt output.

3 Recovery through Rollback

Checkpoint

A **checkpoint** is a saved copy of a process state, including its memory image and the resources assigned to it.

If deadlocks are expected, the system can periodically checkpoint processes. When a deadlock is detected, a process holding a needed resource can be rolled back to an earlier checkpoint before it acquired that resource.

Rollback Idea

$$\boxed{\text{rollback process} \rightarrow \text{release needed resource} \rightarrow \text{restart from checkpoint}}$$

Why old checkpoints are kept

New checkpoints should not overwrite old ones. A sequence of checkpoints gives the system several possible rollback points.

Cost of rollback

All work done after the checkpoint is lost. Any external output produced after the checkpoint may need to be discarded because it may be produced again after restart.

4 Recovery by Killing Processes

Process Killing

The simplest recovery method is to terminate one or more processes until the deadlock is broken.

There are two main choices:

- kill a process that is inside the deadlock cycle,
- kill a process outside the cycle if it holds resources needed by deadlocked processes.

Victim Selection

Where possible, kill a process that can be safely restarted from the beginning with no harmful side effects.

5 Safe and Unsafe Victims

Process type	Safe to kill?	Reason
Compilation	Usually safe	It reads source files and produces an object file. It can be rerun.
Printing job	Sometimes unsafe	Printed output may need to be discarded or restarted carefully.
Database update	Often unsafe	Re-running may apply the update twice and corrupt data.
Transaction with rollback support	Safer	The system can undo partial work before retrying.

Database example

If a process adds 1 to a database field, killing it and running it again may add 1 twice. That changes the data incorrectly.

6 Comparison of Recovery Methods

Method	Main idea	Main problem
Preemption	Take a resource from one process and give it to another.	Many resources cannot be safely preempted.
Rollback	Restart a process from an earlier checkpoint.	Work after the checkpoint is lost.
Killing processes	Terminate one or more processes to release resources.	May lose work or cause incorrect external effects.

7 Worked Example

Problem

Two processes are deadlocked. Process P_1 holds a printer and wants a plotter. Process P_2 holds a plotter and wants a printer. A third process P_3 holds another printer and another plotter but is not deadlocked. How can killing P_3 help?

Solution

Although P_3 is not part of the deadlock cycle, it holds resources that the deadlocked processes need. Killing P_3 releases its printer and plotter. Then one of the deadlocked processes may receive the released resource and continue.

Answer

A recovery victim does not always have to be inside the deadlock cycle. It can be outside the cycle if killing it releases resources needed by the cycle.

8 Choosing a Victim

Good victim

A good victim is cheap to stop, easy to restart, and unlikely to leave corrupted data or duplicated external effects.

Typical factors:

- How much work will be lost?
- Can the process be restarted safely?
- What resources will be released?
- Is the process interactive or batch?
- Could killing it corrupt files, databases, or output?